6.4212 Project Proposal

RhythmBot: Can Robot Play Project SEKAI?

We propose to develop a simulated robotic system capable of playing a continuous rhythm game inspired by *Project SEKAI*. Unlike traditional rhythm games that primarily require discrete button presses, our environment will include **tap, hold, and slide notes**, requiring a robot to execute precisely timed movements while maintaining contact with a playing surface.



Rhythm games provide an interesting testbed for robotic manipulation because they combine spatial precision, temporal coordination, motion planning, and contact-rich control within a structured and easily evaluated environment. In particular, continuous slide notes require the robot to track time-varying trajectories under kinematic and dynamic constraints, while transitions between notes require anticipating future targets and coordinating contact and release.

We will compare the following approaches:

1. Classical motion planning and control: Generate trajectories from known note positions and timing, using inverse kinematics and trajectory tracking.
2. Agentic direct control: Use a language model (e.g. Opus5.5, GPT-6) to generate short sequences of robot actions based on the current environment state.
3. Agentic code-as-policy: Have a language model generate executable programs. We will investigate structured state observations and, if time permits, image-based observations.
4. Behavior cloning: Train a neural policy using demonstrations generated by a classical controller, then evaluate its performance on unseen rhythm charts.

Our primary evaluation metrics will be note completion rate, timing accuracy, continuous trajectory tracking error, contact maintenance, and generalization to unseen note patterns.

Related Work

Our project is inspired by several works on robotic musical performance and robot learning.

RoboPianist (Zakka et al., CoRL 2023) introduces a benchmark for dexterous piano playing with deep reinforcement learning. It demonstrates how musical performance can provide a challenging and measurable testbed for high-dimensional robot control, requiring precise timing and coordination.

PianoMime (Qian et al., CoRL 2024) investigates learning generalist piano-playing policies from human demonstrations. This work motivates our use of behavior cloning and the evaluation of learned policies on previously unseen musical sequences.

GuitarHeroBot, a previous MIT robotic manipulation final project, implements a robot that plays Guitar Hero through timed button presses using predefined trajectories and Drake-based control. Our project builds on the general idea of robotic rhythm-game interaction but extends the task to continuous touchscreen-like trajectories and systematically compares different approaches to generating control policies.

Code as Policies (Liang et al., ICRA 2023) demonstrates that language models can generate executable robot programs incorporating perception, geometric reasoning, and feedback control. Our project investigates whether such programs can handle continuous, time-sensitive interaction and how their performance compares with classical and learned controllers.

References

1. Zakka et al. (2023). *RoboPianist: Dexterous Piano Playing with Deep Reinforcement Learning.* CoRL. https://kzakka.com/robopianist/
2. Qian et al. (2024). *PianoMime: Learning a Generalist, Dexterous Piano Player from Internet Demonstrations.* CoRL. https://pianomime.github.io/
3. *GuitarHeroBot.* MIT 6.4210 Final Project. https://github.com/JRed46/GuitarHeroBot
4. Liang et al. (2023). *Code as Policies: Language Model Programs for Embodied Control.* ICRA. https://code-as-policies.github.io/